DATA ANALYST INTERNSHIP PROJECT

ScreenSense

Kids Screen Time & Health Impact Analysis

Turning Screen Data into Actionable Insights

PROBLEM & OBJECTIVE

Why This Analysis Matters

Children are increasingly exposed to screens — yet there is **no clear visibility** on usage patterns, risk levels, or health impact at scale.

The Problem

- Untracked usage patterns across age groups
- No segmentation of high-risk vs. low-risk users
- Health consequences going unmeasured

Business Objective

- Identify high-risk user segments
- Understand device usage trends
- Analyze behavioral & health impact
- Provide actionable recommendations

Dataset & Key Metrics Snapshot

Dataset Profile

Total Records: 9,668

Type: Behavioral + Health Data

Key Features Tracked:

- Age Group & Gender
- Device Type
- Daily Screen Time (hrs)
- Usage Type: Educational vs. Recreational
- Health Impact Indicators

4.37

Avg Screen Time

Hours per day across all users

91.75%

Exceed Safe Limit

Majority surpass recommended screen time

85.77%

High Risk Users

Classified as high-risk based on usage patterns

Key Insight: The overwhelming majority of kids in this dataset exceed safe screen limits — this is a systemic issue, not an isolated one.

Screen Time & Device Usage Insights

Screen Time Trends

Age-Driven Usage

Screen time increases consistently with age. Peak usage is observed in the **17–18 age group**.

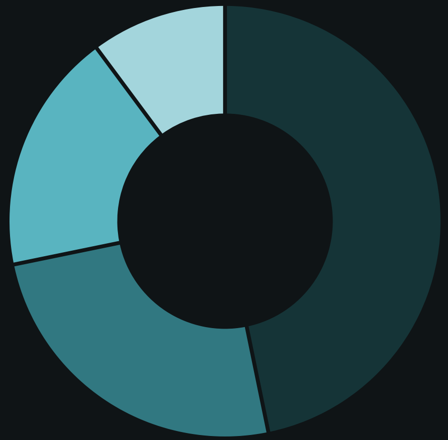
Gender Is Not a Factor

Minimal gender difference found. **Screen usage is age-driven, not gender-driven.**

Early Exposure Is Common

Even younger age groups show significant daily screen time well above recommended levels.

Device Distribution



Insight: Smartphones dominate at ~47%. Usage shifts from shared (TV) to personal (smartphone) as age increases.

Behavior Patterns & Risk Levels

Usage Intent by Age

Young Kids

Higher proportion of educational content

Tweens

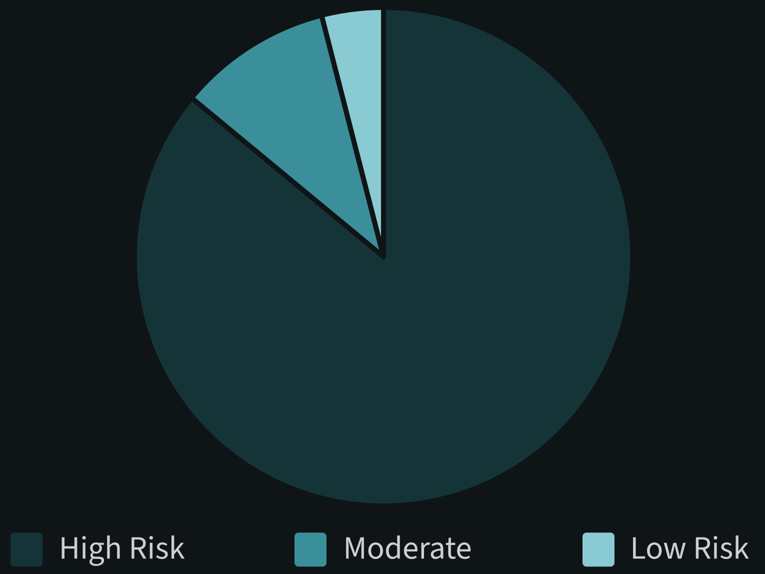
Balanced mix shifting toward recreation

Teenagers

~75–80% recreational usage dominates

Insight: Usage intent shifts with age — teenagers are almost entirely entertainment-driven.

Risk Level Distribution



Insight: Screen overexposure is **widespread, not isolated** — 85%+ of users fall in the high-risk category.

HEALTH & GEOGRAPHIC INSIGHTS

Health Impact & Urban vs. Rural Divide

Top Health Issues Linked to Screen Overuse

Poor Sleep

Most common health complaint across all age groups

Eye Strain

Directly correlated with device type and duration

Anxiety

Stronger in teenagers with high recreational usage

Critical Insight: Health risks increase significantly once screen time crosses the recommended daily limit.

Urban vs. Rural Behavior

Urban Users

Higher overall screen time. Greater access to personal devices. More recreational & entertainment-oriented usage.

Rural Users

Lower usage on average. Slightly higher proportion of educational content. Limited device variety.

Insight: Lifestyle and device accessibility are the primary drivers of screen habits — not age alone.

User Segmentation Framework

🔴 High Risk Heavy Users

Who: Primarily teenagers (13–18), urban, smartphone-dominant

- Screen time far exceeds daily limits
- Recreational usage ~80%+
- Highest reported health issues
- Immediate intervention needed

🟡 Moderate Users

Who: Tweens (9–12), mixed urban/rural, tablet & TV users

- Near-limit screen time
- Balanced educational-recreational mix
- Some sleep and eye strain reported
- Preventive action recommended

🟢 Low Risk Users

Who: Younger children (5–8), rural, supervised usage

- Within safe screen time limits
- Predominantly educational content
- Minimal health impact reported
- Monitor as they grow older

What the Data Is Telling Us

91% of kids exceed recommended screen time limits

This is not an outlier problem — it is the norm across the entire dataset of 9,668 records.

Smartphones are the single biggest driver of overuse

Personal device ownership correlates directly with higher and less supervised usage.

Teenagers represent the highest-risk segment

Peak usage at age 17–18 with recreational dominance and the highest health impact scores.

Health risks spike sharply beyond the safe limit

Poor sleep, eye strain, and anxiety are strongly correlated with exceeding the daily threshold.

Recreational usage dominates — educational content is underutilized

75–80% of screen time is non-educational, especially among older children and urban users.

RECOMMENDATIONS & IMPACT

Actionable Recommendations & Business Impact

Recommendations

Introduce Screen Time Limits

Enforce daily caps by age group, especially for smartphones and teenagers.

Promote Educational Content

Curate and surface educational apps for younger age groups to shift usage intent.

Encourage Device-Free Routines

Support bedtime and mealtime screen-free policies to improve sleep and wellbeing.

Monitor Smartphone Usage

Focus intervention on smartphone behavior — the dominant and most unregulated channel.

Real-World Impact

Parents

Helps families manage and reduce harmful screen habits with data-backed guidance.

Schools

Supports digital wellness awareness programs targeted at the highest-risk age groups.

Policy Makers

Provides empirical evidence for legislation around screen time guidelines for children.

TOOLS USED

Power BI Interactive dashboards & visual storytelling	**Data Cleaning** Structured preparation of 9,668 behavioral records
Data Visualization Clear, insight-driven visual design throughout	**Analytical Thinking** Segment-based storytelling and pattern recognition

Screen time is a growing crisis. Data reveals the patterns. Action reduces the risk.

Thank you for your time — **Questions?**